2.1 'FULL-CORE' VVER-440 PIN POWER DISTRIBUTION CALCULATION BENCHMARK

Václav Krýsl, Pavel Mikoláš, Daniel Sprinzl, Jiří Švarný
(Corresponding author: pavel.mikolas@skoda-js.cz)
ŠKODA JS a. s.

ABSTRACT

The 'Full-Core' benchmark [1] is a 2D calculation benchmark based on the VVER-440 reactor core cold state geometry with taking into the account the geometry of explicit radial reflector. The main task of this benchmark is to test the pin by pin power distribution in fuel assemblies that are placed mainly at the border of the VVER-440 core. Because value of F_{dH} is not directly measured by the core monitoring system, a decision about a proposal of benchmark of this kind has been made. In this contribution we define the 'Full-Core' benchmark and we present the preliminary reference Monte Carlo calculation results.

1. INTRODUCTION

The idea of "'Full-Core' calculation benchmark for VVER-440 reactor" was motivated by earlier MIDICORE VVER-1000 calculation benchmark [2] mainly due to the validation of fuel pin power distribution of MOBY-DICK macro-code. We want to test the pin by pin power distribution mainly in the FA's near the border of active zone (AZ) in this benchmark.

An as much as possible independent method of benchmark solution is needed to obtain the reference solution. Such a suitable solution method might be the Monte Carlo method. Despite the time consumption of Monte Carlo based calculations the whole AZ (respectively a given segment of its symmetry described later) is calculated to avoid troubles with macro-code calculation of smaller parts of AZ (as in the case of MIDICORE VVER-1000 benchmark). Apart from the reason mentioned in abstract we believe that this benchmark might be useful for example in further upgrade of methodic of calculation with explicit radial reflector modeled as a diffusion region.

2. “FULL-CORE” BENCHMARK PROPOSAL

Benchmark specification: In general the Full-Core is a 2D benchmark based on cold state geometry. Due to the 30 degree reflection symmetry of both loading pattern from Fig. 1a and the radial reflector this benchmark is restricted only on a 30 degree segment of VVER-440 AZ as denoted in Fig. 1b (the red curves) in the model of horizontal cut of 1/6 VVER-440 core. The outer edge of the reactor pressure vessel presents the outer boundary of the model in the radial direction. Both the core and radial reflector structure in axial direction is treated like infinite homogeneous.

The boundary conditions of the Full-Core model are specified as follows. The reflection boundary conditions are used on the azimuthal edges in Fig. 1a while the total absorption boundary condition (leakage to the vacuum) is used on the cylindrical outer boundary. The reflection boundary conditions are used also in axial directions.

The dimensions of the model of the radial reflector structure are based on the available technical documentation [5]. Unfortunately due to the unreadable of the drawing in this small format the drawing is not included here but the dimensions of the cylindrical features in the radial reflector model are described in Fig. 1b and in table 1. The thickness of ‘vygorodka’ (internal edged metal sheet) is 8 mm and the gap between the ‘vygorodka’ and the FA model hexagon with pitch 147 mm is 1.5 mm (that means 2.5 mm water gap between ‘vygorodka’ and the border FA shroud).

The distance of 147 mm between the centers of two neighboring FA’s is treated in the cold state geometry. In this sense the horizontal cut of the core may be covered by a net of hexagons and by this way so called reflector FA’s may be modeled in the macro-codes. The cartogram of the Full-Core model can be found in Fig 1a. The names of FA’s correspond to their average enrichment and their details are defined in Figs. 2a-2c and in table 1. The control FA’s are of course treated as fully inserted in the AZ. The central tubes of all FA’s are filled with moderator. The both fuel and non-fuel material compositions (including moderator) are defined in tables 2, 3 and 4.

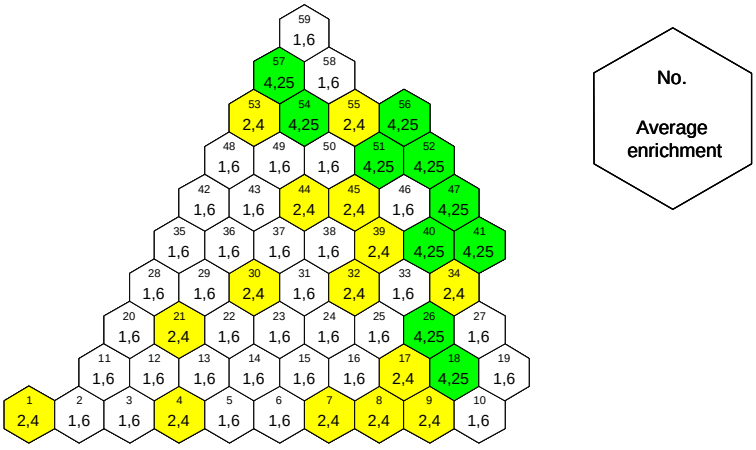


Fig 1a: Loading pattern with 30 degree reflection symmetry composed of three different types of fresh FA marked by different colors (FA design based on [3] and [4]).

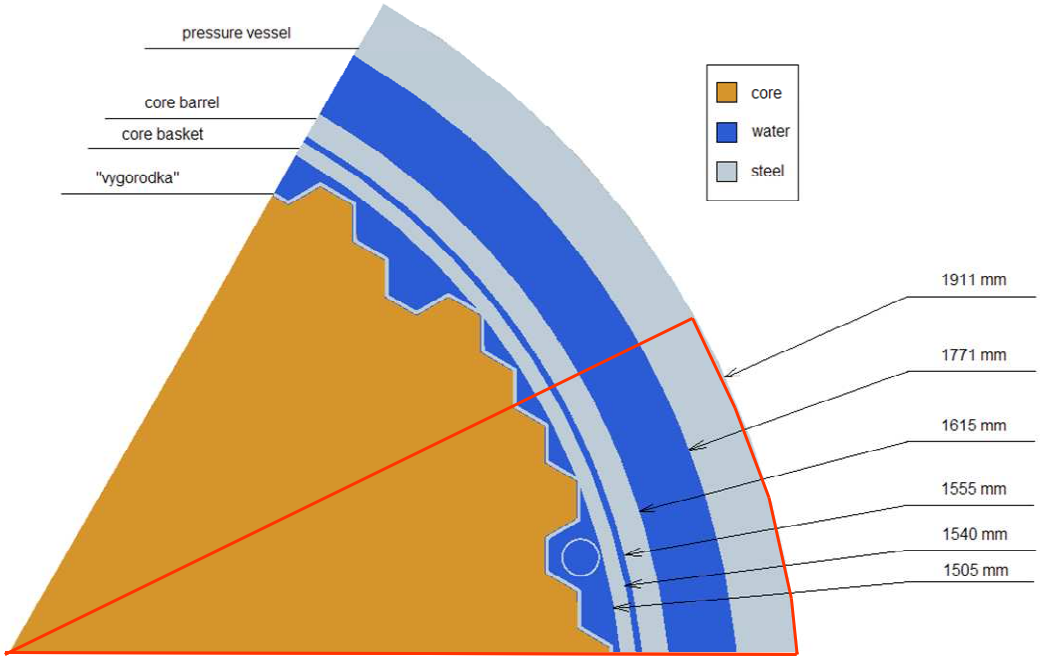


Fig. 1b: The Full-Core calculation model (denoted by the red line) as a segment of horizontal cut of the VVER-440 core. The steel tube between the "vygorodka" and the core basket drawn in the picture is neglected in the calculations.

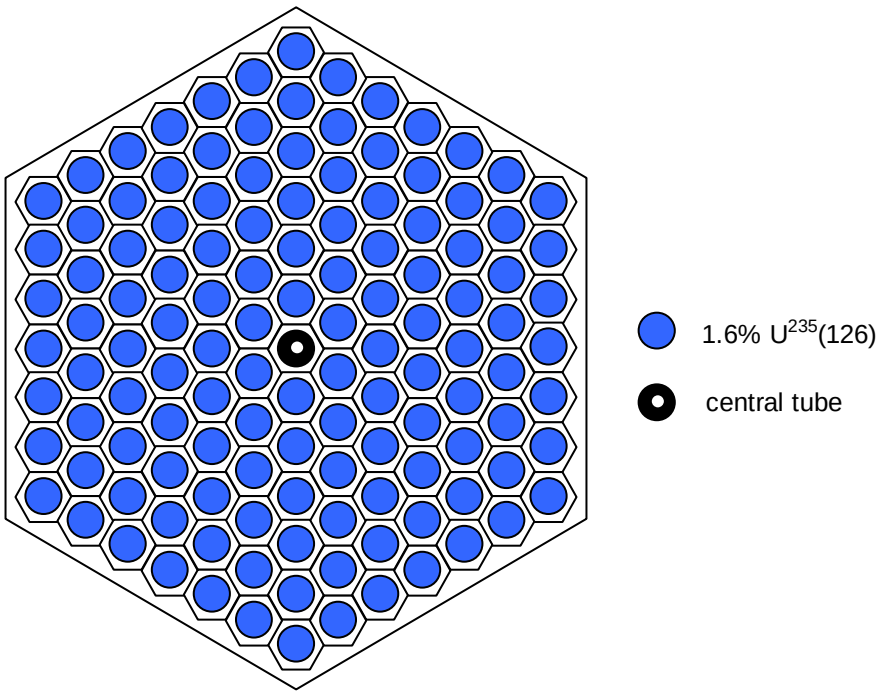


Fig 2a: FA of type 1.6

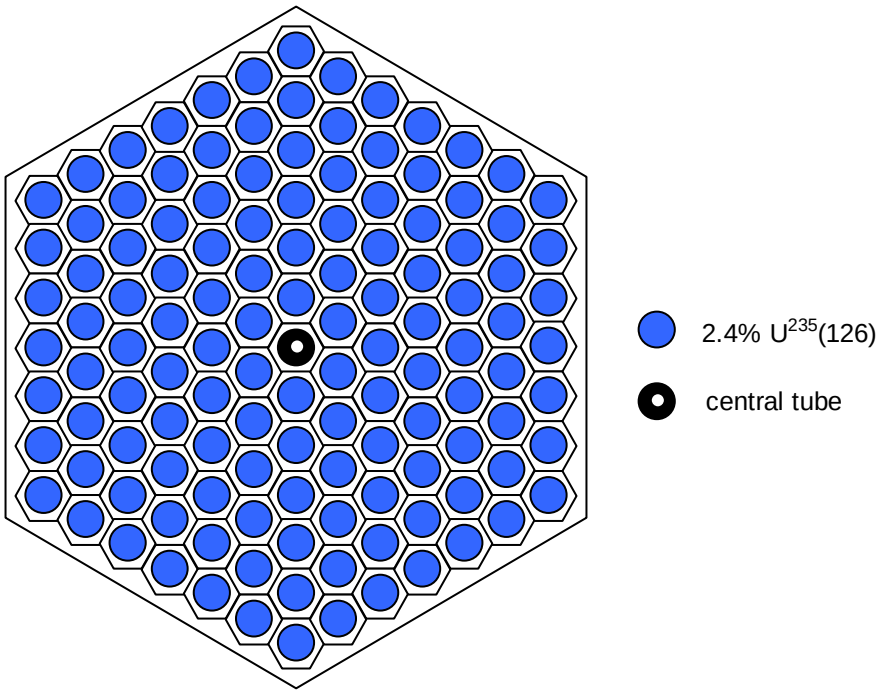


Fig 2a: FA of type 2.4

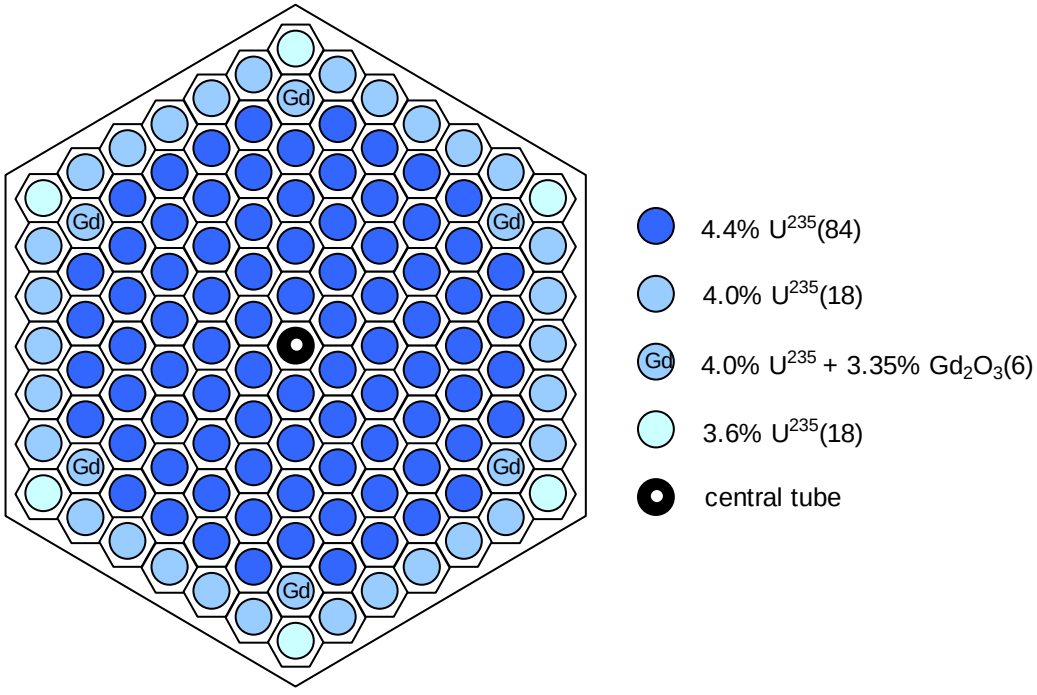


Fig 2a: FA of type 4.25

Table 1: Dimension and fuel specification of Full-Core model.

Parameter	**Quantity**
FA pitch in AZ, [cm]	14.7
FA nominal outer diameter, [cm]	14.5
FA number of fuel pins	126
FA number of central tubes	1
FA number of grids	0
FA shroud thickness, [cm]	0.15
FA shroud height, [cm]	infinity
FA shroud material	E-125
Central tube material	E-110
Central tube outer diameter, [cm]	1.03
Central tube inner diameter, [cm]	0.88
Fuel pins spacing, [cm]	1.23
Fuel pins height, [cm]	infinity
Fuel pins cladding material (TVEL, TVEG)	E-110
Fuel pins cladding outer diameter (TVEL, TVEG), [cm]	0.91
Fuel pins cladding inner diameter (TVEL, TVEG), [cm]	0.773
Fuel pellet material (TVEL)	UO_2
Fuel pellet material (TVEG)	$UO_2 + Gd_2O_3$
Fuel pellet outer diameter, [cm]	0.76
Fuel pellet inner diameter, [cm]	0.12
TVEG fuel pellet Gd_2O_3 weight ratio, [%w Gd_2O_3]	3.35
TVEG fuel pellet U^{235} enrichment, [%w U^{235}]	4.0
Temperature of all materials, [K]	543.15
Pressure, [MPa]	12.3
FA 1.6 (Fig. 2a)	
The number of TVEL fuel pins with enrichment 1.6 %w U^{235}	126
FA 2.4 (Fig. 2b)	
The number of TVEL fuel pins with enrichment 2.4 %w U^{235}	126
FA 4.25 (Fig. 2c) (profiled)	
The number of TVEL fuel pins with enrichment 4.4 %w U^{235}	84
The number of TVEL fuel pins with enrichment 4.0 %w U^{235}	18
The number of TVEL fuel pins with enrichment 3.6 %w U^{235}	18
The number of TVEG fuel pins with enrichment 4.0 %w U^{235} (+ 5.0 %w of Gd_2O_3)	6
Dimensions	
Core barrel outer radius, [cm]	191.1
Core barrel inner radius, [cm]	177.1
Core barrel outer radius, [cm]	161.5
Core barrel inner radius, [cm]	155.5
Core basket outer radius, [cm]	154.0
Core basket inner radius, [cm]	150.5
'Vygorodka' thickness, [cm]	0.8
Water gap thickness, [cm] (between vygorodka and the edge of model FA hexagon with pitch 14.7 cm)	1.5

Table 2: Basic non-fuel material specification of Full-Core model.

Parameter	Quantity
Material E-110	
Density, [g/cm^3]	6.55
Composition, [%w] Zr(98.78)+Nb(1.00)+O(0.10)+Fe(0.05)+Ni(0.02)+Cr(0.02)+C(0.02)+Hf(0.01)	
Material E-125	
Density, [g/cm^3]	6.55
Composition, [%w] Zr(97.28)+Nb(2.50)+O(0.10)+Fe(0.05)+Ni(0.02)+Cr(0.02)+C(0.02)+Hf(0.01)	
Pressure vessel, core barrel, core basket, 'vygorodka'	
Material	steel
Density, [g /cm^3]	8.0
Composition, [%w] Fe(68.535)+Cr(18.000)+Ni(10.000)+Mn(2.000)+Si(0.800)+Ti(0.550)+C(0.080) +P(0.035)	
Moderator	
Content of absorbing material in moderator, [g of H_3BO_3/kg of mixture]	3.0
Density, [g /cm^3] (for temperature 543.15 K and pressure 12.3 MPa)	0,777537
Composition, [%w] H_2O (99.70)+H_3BO_3 (0.30)	

Table 3: The detail fuel material specification of Full-Core model.

Density of nuclei for different fuel pellets, [1/cm^3]						
Composition/ enrichment	1,6	2,4	3,6	4,0	4,4	4,0 with Gd
O	4,64026E+22	4,64069E+22	4,64132E+22	4,64153E+22	4,64174E+22	4,65970E+22
U^{234}	3,02003E+18	4,52999E+18	6,79486E+18	7,54980E+18	8,30473E+18	7,29688E+18
U^{235}	3,75892E+20	5,63831E+20	8,45731E+20	9,39696E+20	1,03366E+21	9,08216E+20
U^{238}	2,28224E+22	2,26351E+22	2,23541E+22	2,22604E+22	2,21668E+22	2,15147E+22
Gd^{154}	-	-	-	-	-	2,52388E+19
Gd^{155}	-	-	-	-	-	1,71346E+20
Gd^{156}	-	-	-	-	-	2,36990E+20
Gd^{157}	-	-	-	-	-	1,81187E+20
Gd^{158}	-	-	-	-	-	2,87583E+20

Table 4: The detail non-fuel material specification of Full-Core model.

Density of nuclei for non-fuel materials, [1/cm^3]		
Composition/material	**E-110**	**E-125**
Zr	4,27122E+22	4,20636E+22
Nb	4,24582E+20	1,06145E+21
O	2,46677E+20	2,46677E+20
Fe	3,53172E+19	3,53172E+19
C	6,56812E+19	6,56812E+19

Cr	1,51724E+19	1,51724E+19
Ni	1,34410E+19	1,34410E+19
Hf^{174}	3,62837E+15	3,62837E+15
Hf^{176}	1,16806E+17	1,16806E+17
Hf^{177}	4,13756E+17	4,13756E+17
Hf^{178}	6,00743E+17	6,00743E+17
Hf^{179}	3,03083E+17	3,03083E+17
Hf^{180}	7,72022E+17	7,72022E+17
Composition/material	**Pressure vessel, core barrel, core basket, 'vygorodka'**	
Fe	5,91258E+22	
Cr	1,66781E+22	
Ni	8,20823E+21	
Mn	1,75387E+21	
Si	1,37231E+21	
Ti	5,53547E+20	
C	3,20885E+20	
P	5,44392E+19	
Composition/material	**Moderator**	
H	5,19820E+22	
O	2,59909E+22	
B^{10}	4,53447E+18	
B^{11}	1,82500E+19	

Although all the dimensions are in the cold state geometry the temperature of FA's and moderator is 543.15 K. The pressure is 12.3 MPa and corresponding density of moderator 0.777537 g/cm^3.

Benchmark's objective: The main issue of this Full-Core benchmark is to give some reference for validation of accuracy of pin by pin power distribution mainly at the periphery of the VVER-440 AZ calculated by standard codes using few-group diffusion approximation. The Full-Core benchmark objectives are:

1) K_{eff};
2) Pin by pin power distribution in FA
3) Integral fission power of FA's.

Benchmark suggestions: The first suggestion is to take into account at least one row of the "reflector" FA's in the macro-code calculation. Second suggestion deals with the norm of the results. The integral fission power from FA's should be normalized in such way that the mean value of all 349 FA's in 360 degree symmetry equals 1. The pin by pin power distribution should be normalized in such way that the mean value of relative pin powers equals 1 in each FA.

3. PRELIMINARY REFERENCE MONTE CARLO CALCULATION RESULTS

All details of modeling in Monte Carlo calculation follow information presented in chapter 2 of this paper.

Monte Carlo code and libraries used: All presented Monte Carlo calculations have been performed with MCNP-4C code [6]. The ENDF/B6 library was chosen as the basic data library. The calculations have been performed for the temperature 543.15 K in option tmp and with thermalization lwtr.04t.

In development of benchmark reference values by Monte Carlo code the question about precision and accuracy of power distribution calculation should be in our focus.

The Monte Carlo precision analysis: The main factor affecting precision of Monte Carlo power distribution calculation are number of neutrons per generation (n.), number of generations (gen.) and in our case also positions of *ksrc* sources (we have positioned *ksrc* sources in each FA).

The presented informative results show calculation with the 200 mil. nps and 200 000 n. per generation:

a) K_{eff} = 1.09216 σ = 0.00004

b) Pin by pin power distribution in FA 40 (Fig. 3a), 34 (Fig. 3b), 27 (Fig. 3c) (numerating acc. to Fig. 1a).

c) Relative FA power with relative errors from Monte Carlo calculation Fig. 3d.

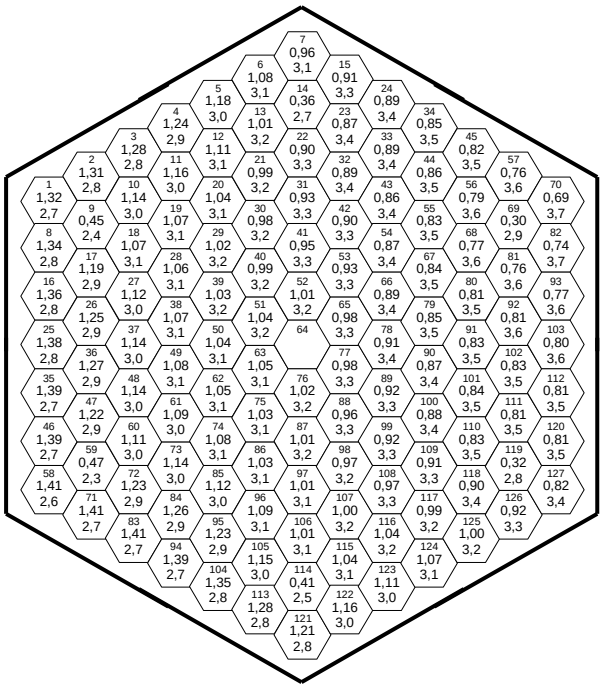


Fig 3a: Relative pin power distribution for FA No. 40 (4.25) calculated by MCNP. The upper number in each cell is the pin number, the bottom number is the relative error [10^{-3}].

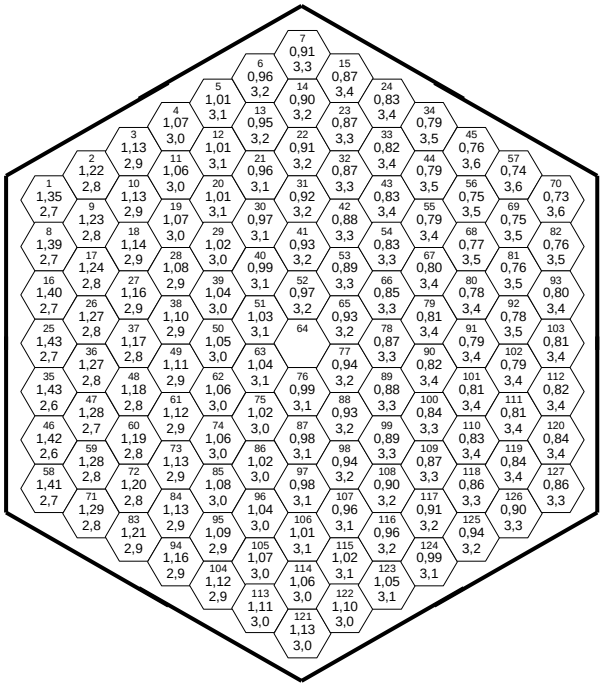


Fig. 3b: Relative pin power distribution for FA No. 34 (2.4) calculated by MCNP. The upper number in each cell is the pin number, the bottom number is the relative error [10^{-3}].

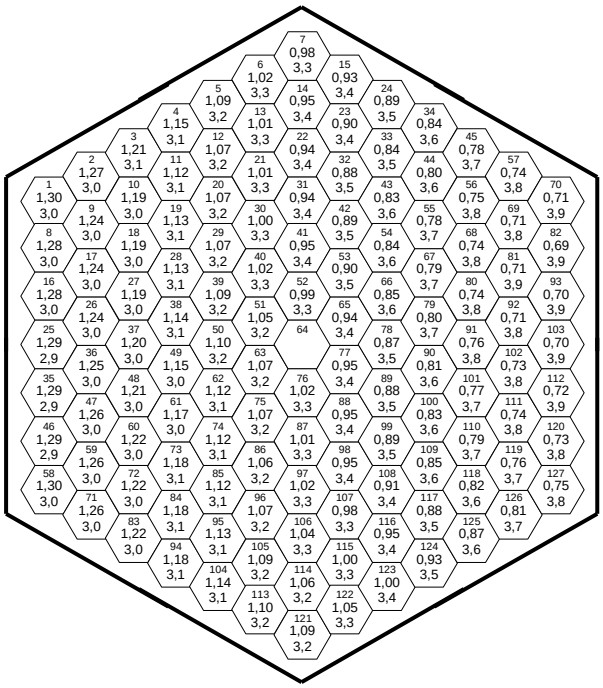


Fig 3c: Relative pin power distribution for FA No. 27 (1.6) calculated by MCNP. The upper number in each cell is the pin number, the bottom number is the relative error [10^{-3}].

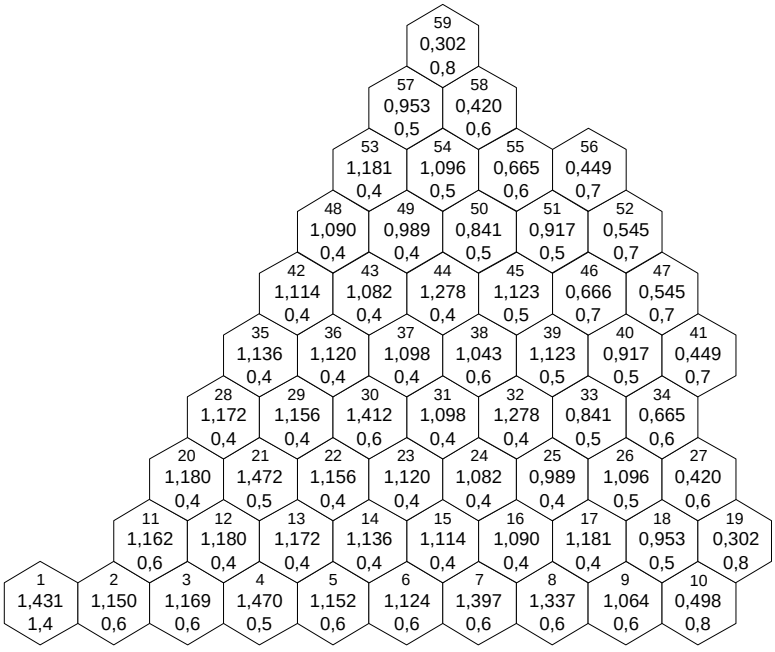


Fig 3d: Relative FA power distribution for the Full-Core model calculated by MCNP. The upper numbers are the FA's numbering, the bottom numbers are the relative errors [10^{-3}].

4. CONCLUSION

We have proposed a benchmark 'Full-Core' for VVER-440 to test the pin by pin power distribution of macro-codes. We have also presented preliminary results from Monte Carlo reference solution. However, the presented MCNP results are only informative mainly due to the (at this moment) low nps number that is not sufficient for such a large simulated structure. Therefore the reference solution will come later. From this point of view of MCNP reference solution there would be very useful to confirm the results by independent Monte Carlo calculations. Therefore Monte Carlo results as well as the macro-code results are welcomed.

LIST OF NOMENCLATURE

AZ	-	active zone
FA	-	fuel assembly
MCNP	-	Monte Carlo calculation code [6]
Full-Core	-	name of the benchmark
TVEG	-	fuel pin with burnable absorber
TVEL	-	fuel pin without burnable absorber
%w	-	weight percent

REFERENCES

[1] V. Krýsl, P. Mikoláš, D. Sprinzl, J. Švarný: *'Full-Core' VVER-440 pin power distribution benchmark proposal*. AER Working Groups A&B, Plzeň, May 10-11, 2011, Czech Republic.

[2] V. Krýsl, P. Mikoláš, D. Sprinzl, J. Švarný: *'MIDICORE' VVER-1000 core periphery power distribution benchmark proposal*. Proceedings of the twentieth Symposium of AER, 2010.

[3] ČEZ, a.s. – TVEL, The set of fuel assemblies for VVER-440, Contract technical agreement, 2007.

[4] SE, a.s.– TVEL, Nuclear Fuel Contract, 2009.

[5] ŠKODA JS, a.s., technical documentation for VVER-440 V-213, drawings Ae004481, Ae005098.

[6] J.F. Briesmeister: *MCNP manual*, LA-12625-M, Version 4C.